

Breathing & Exchange of Gases

Respiratory Organs:-

The mechanisms of breathing differ among animal groups depending largely on their habitats and levels of organization. Lower invertebrates, including sponges, coelenterates, and flatworms, exchange and by simple diffusion across their entire body surface. Earthworms use their moist cuticle. Insects possess a network of tubes known as tracheal tubes to transport atmospheric air internally. Most aquatic arthropods and molluscs use vascularized structures called gills (branchial respiration), while terrestrial forms use vascularized bags called lungs (pulmonary respiration) for gas exchange. Among vertebrates, fishes use gills, while amphibians, reptiles, birds, and mammals respire through lungs. Amphibians, such as frogs, can also respire through their moist skin (cutaneous respiration).

Human Respiratory System Anatomy

1. Air Passage: We have a pair of external nostrils opening above the upper lips, leading to a nasal chamber through the nasal passage. The nasal chamber opens into the pharynx, which serves as a common passage for food and air. The pharynx opens into the trachea through the region of the larynx. The larynx is a cartilaginous box, often called the sound box, because it assists in sound production. During swallowing, a thin, elastic cartilaginous flap called the epiglottis covers the glottis to prevent food entry into the larynx.

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2. **Trachea and Lungs:** The trachea is a straight tube extending up to the mid-thoracic cavity, where it divides at the level of the 5th thoracic vertebra into the right and left primary bronchi. Each bronchus repeatedly divides into secondary and tertiary bronchi, followed by bronchioles, and ending in very thin terminal bronchioles. The trachea, bronchi, and initial bronchioles are supported by incomplete cartilaginous rings.

3. **Alveoli:** Each terminal bronchiole culminates in numerous very thin, irregular-walled, and highly vascularized bag-like structures called alveoli. The branching network of bronchi, bronchioles, and alveoli constitutes the lungs.

4. **Pleura:** The two lungs are covered by a double-layered membrane called the pleura, which contains pleural fluid between the layers to reduce friction on the lung surface. The outer pleural membrane is in close contact with the thoracic lining, while the inner pleural membrane touches the lung surface.

5. Functional Parts:

- The conducting part runs from the external nostrils up to the terminal bronchioles. This part transports atmospheric air to the alveoli, clears it of foreign particles, humidifies it, and brings the air to body temperature.

- The respiratory or exchange part consists of the alveoli and their ducts. This is the site of the actual diffusion of and between the blood and atmospheric air.

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6. Thoracic Chamber: The lungs reside within the thoracic chamber, which is an anatomically air-tight enclosure. This chamber is defined dorsally by the vertebral column, ventrally by the sternum, laterally by the ribs, and inferiorly by the dome-shaped diaphragm. The setup ensures that any volume change in the thoracic cavity is reflected in the pulmonary volume, which is crucial for breathing since pulmonary volume cannot be altered directly.

Mechanism of Breathing:-

Breathing involves inspiration (drawing atmospheric air in) and expiration (releasing alveolar air out). Air movement relies on creating a pressure gradient between the lungs (intra-pulmonary pressure) and the atmosphere.

- Inspiration occurs when the intra-pulmonary pressure is lower (negative pressure) than the atmospheric pressure.
- Expiration occurs when the intra-pulmonary pressure is higher than the atmospheric pressure.
- The diaphragm and specialized external and internal intercostal muscles generate these gradients.

The Process:

1. Inspiration: Initiated by the contraction of the diaphragm, increasing the volume of the thoracic chamber in the antero-posterior axis. Contraction of the external intercostal muscles lifts the ribs and sternum, increasing the thoracic chamber volume in the dorso-ventral axis. This overall increase in thoracic volume causes a similar increase in pulmonary volume. The resultant drop in intra-pulmonary

The resulting drop in intra-pulmonary pressure below atmospheric pressure forces air into the lungs.

2. **Expiration:** Involves the relaxation of the diaphragm and intercostal muscles, returning the sternum and diaphragm to their normal positions, thus reducing both thoracic and pulmonary volume. This leads to an increase in intra-pulmonary pressure slightly above atmospheric pressure, causing the air to be expelled. The strength of both inspiration and expiration can be increased using additional abdominal muscles. A healthy human typically breathes 12-16 times per minute. The volume of air involved in breathing can be estimated using a spirometer, which is helpful for the clinical assessment of pulmonary functions.

Respiratory Volumes and Capacities:-

Pulmonary capacities are derived by adding volumes and are used in clinical diagnosis.

- **Tidal Volume (TV):** The volume of air inspired or expired during a normal respiration, approximately 500 mL. A healthy man can inspire or expire around 6000 to 8000 mL of air per minute.
- **Inspiratory Reserve Volume (IRV):** The maximum volume of air a person can inspire by a forcible inspiration, averaging 2500 mL to 3000 mL.
- **Expiratory Reserve Volume (ERV):** The maximum volume of air a person can expire by a forcible expiration, averaging 1000 mL to 1100 mL.
- **Residual Volume (RV):** The volume of air remaining in the

- **Residual Volume (RV):** The volume of air remaining in the lungs even after a forcible expiration, averaging 1100 mL to 1200 mL.

Capacities:-

- **Inspiratory Capacity (IC):** Total volume of air a person can inspire after a normal expiration ($TV + IRV$).
- **Expiratory Capacity (EC):** Total volume of air a person can expire after a normal inspiration ($TV + ERV$).
- **Functional Residual Capacity (FRC):** Volume of air that remains in the lungs after a normal expiration ($ERV + RV$).
- **Vital Capacity (VC):** The maximum volume of air a person can breathe in after a forced expiration ($ERV + TV + IRV$), which is also the maximum volume of air a person can breathe out after a forced inspiration.
- **Total Lung Capacity (TLC):** Total volume of air accommodated in the lungs at the end of a forced inspiration ($RV + ERV + TV + IRV$), equivalent to Vital Capacity + Residual Volume.

III. Exchange and Transport of Gases:-

Gas Exchange Principles

The primary sites of gas exchange are the alveoli, though exchange also occurs between the blood and tissues. Gases are exchanged by simple diffusion driven mainly by the pressure/concentration gradient. Factors affecting the diffusion rate include the solubility of the gases and the thickness of the diffusion membranes.

The pressure contributed by an individual gas in a mixture is called its partial pressure (or).

- A concentration gradient for exists from the alveoli to the blood, and from the blood to the tissues.
- A gradient for exists in the opposite direction: from the tissues to the blood, and from the blood to the alveoli.
- The solubility of CO_2 is 20-25 times higher than that of O_2 . Consequently, the amount of that can diffuse per unit difference in partial pressure is much higher compared to .

Diffusion Membrane: The diffusion membrane is exceptionally thin, totaling much less than a millimeter in thickness. It is composed of three main layers:

1. The thin squamous epithelium of the alveoli.
2. The endothelium of the alveolar capillaries.
3. The basement substance situated between the two, which consists of a thin basement membrane supporting the squamous epithelium and the basement membrane surrounding the capillary endothelial cells. These factors collectively ensure favorable conditions for the diffusion of from alveoli to tissues and from tissues to alveoli.

Blood transports and .

- 97% of O_2 is transported by Red Blood Cells (RBCs).
- The remaining 3% of O_2 is carried dissolved in the plasma.

Haemoglobin Binding:

- Haemoglobin is a red-colored, iron-containing pigment found in RBCs.
- binds reversibly with haemoglobin to form oxyhaemoglobin.
- Each haemoglobin molecule can carry up to four molecules.
- Binding is primarily related to the partial pressure of O_2 , but factors like pH , hydrogen ion $[H^+]$ concentration, and temperature also interfere.
- The relationship between pO_2 and the percentage saturation of haemoglobin with O_2 yields a sigmoid curve called the Oxygen Dissociation Curve. This curve is vital for studying how factors like pH and concentration affect binding.
- In the alveoli (high pO_2 , low pCO_2 , lesser concentration, lower temperature), conditions favor the formation of oxyhaemoglobin.
- In the tissues (low pO_2 , high pCO_2 , high concentration, higher temperature), conditions favor the dissociation of oxygen from oxyhaemoglobin.
- Under normal physiological conditions, every 100 ml of oxygenated blood delivers approximately 5 ml of O_2 to the tissues.

Transport of Carbon Dioxide CO_2 is transported in three ways:

- 70% is carried as bicarbonate HCO_3^- .
- 20-25% is carried by haemoglobin as carbamino-haemoglobin.
- 7% is carried dissolved in the plasma.

Carbamino-haemoglobin:

- The binding of to haemoglobin is related to , and significantly affects this bond.
- In the tissues (high , low), more binds.
- In the alveoli (low , high), dissociates from carbamino-haemoglobin and is delivered out.

Bicarbonate Formation:

- The enzyme carbonic anhydrase is present in very high concentration in RBCs and in minute quantities in the plasma.
- This enzyme facilitates the rapid and reversible reaction in both directions:
- At the tissue site, the high (due to catabolism) causes to diffuse into the blood (RBCs and plasma) and form bicarbonate ions (HCO_3^-) and hydrogen ions (H^+).
- At the alveolar site, the low causes the reaction to reverse, leading to the formation and release of and water (H_2O).
- Every 100 ml of deoxygenated blood delivers approximately 4 ml of CO_2 to the alveoli.

IV. Regulation of Respiration and Disorders:-

Regulation of Respiration

The body utilizes the neural system to maintain and moderate the respiratory rhythm to suit the demands of the body tissues.

1. Respiratory Rhythm Centre: Located in the medulla region of the brain, this is the primary center responsible for respiratory regulation.

2. Pneumotaxic Centre: Located in the pons region of the brain. Neural signals from this center can moderate the rhythm center's functions, specifically by reducing the duration of inspiration, thereby altering the respiratory rate.

3. Chemosensitive Area: Situated adjacent to the rhythm center in the medulla, this area is highly sensitive to CO_2 and hydrogen ions H^+ . An increase in these substances activates this area, which then signals the rhythm center to make necessary adjustments to eliminate the excess and .

4. Peripheral Receptors: Receptors associated with the aortic arch and carotid artery can also recognize changes in and concentration and transmit signals to the rhythm center for remedial actions. The role of oxygen in regulating respiratory rhythm is considered quite insignificant.

Disorders of the Respiratory System

1. Asthma: Characterized by difficulty in breathing that causes wheezing, typically due to the inflammation of the bronchi and bronchioles.

2. Emphysema: A chronic disorder where the alveolar walls are damaged, leading to a decreased respiratory surface. A major cause of emphysema is cigarette smoking.

3. Occupational Respiratory Disorders: These occur in industries like grinding or stone-breaking where excessive dust production overwhelms the body's defense mechanisms.